

Mindtree Internship Report

An internship report

Submitted in partial fulfillment of the requirements for the award of
degree of

Master of Computer Application

Submitted to

LOVELY PROFESSIONAL

UNIVERSITY PHAGWARA, PUNJAB



From 02/22/2023 to 05/31/2023

SUBMITTED BY

Atul Kumar

12102801

Submitted to:

Dr. Ajay Rastogi

**Professor
(Placement Mentor)**

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To whom so ever it may concern

This is to certify that **Atul Kumar, 12102801** from Lovely Professional University, Phagwara, Punjab, has worked as an intern in **Mindtree Limited** on “**Dot Net**” under my supervision from **February 2023** to **May 2023**. It is further stated that the work carried out by the student is a record of original work to the best of my knowledge for the partial fulfillment of the requirements for the award of the degree, **Master of Computer Application**.

Dr. Ajay Rastogi

Professor & Placement Mentor

June 23rd, 2023

To whom so ever it may concern

I, **Atul Kumar, 12102801** hereby declare that the work done by me on “**Dot Net**” from **February 2023** to **May 2023**, under the supervision of **Mindtree Limited** and **Dr. Ajay Rastogi, Professor**, Lovely professional University, Phagwara, Punjab, is a record of original work for the partial fulfillment of the requirements for the award of the degree, **Master of Computer Application**.

Atul Kumar (12102801)

June 23rd, 2023

Course Certificate



Acknowledgement

The internship opportunity I had with MINDTREE was a great chance for learning and professional development. Therefore, I consider myself as a very lucky individual as I was provided with an opportunity to be a part of it. I am also grateful for having a chance to meet so many wonderful people and professionals who led me through this internship period.

Bearing in mind previous I am using this opportunity to express my deepest gratitude and special thanks to the Ignite team who in spite of being extraordinarily busy with their duties, took time out to hear, guide and keep me on the correct path and allowing me to carry out my course at their esteemed organization and extending during the training.

I express my deepest thanks to Dr. Ajay Rastogi, Professor & Placement Mentor for taking part in useful decision & giving necessary advice and guidance and arranged all facilities to make life easier. I choose this moment to acknowledge his contribution gratefully.

Atul Kumar

12102801

June 23rd, 2023

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List of Abbreviations:

LBJ - Learning Before Joining
SME - Subject Matter Expert
VILT - Virtual Interact Let's Talk
CEO - Chief Executive Officer
CRM - Customer Relationship Management
KPI - Key Performance Indicator
ROI - Return on Investment
MIS - Management Information System
SWOT - Strengths, Weaknesses, Opportunities, Threats
HR - Human Resources
IT - Information Technology
SOP - Standard Operating Procedure
R&D - Research and Development
FAQ - Frequently Asked Questions
NPS - Net Promoter Score
QC - Quality Control
SLA - Service Level Agreement
CRM - Customer Relationship Management
ERP - Enterprise Resource Planning

Chapter 1: Introduction of The Company

Company Profile Mindtree (Now LTIMindtree)



Mindtree Ltd is an Indian multinational information technology services and consulting company, headquartered in Bangalore. It is a part of the Larsen & Toubro Group. Founded in 1999, the company employs over 23,000 employees and has an annual revenue of ₹7839.9 crore (US\$1.1 billion).

The company deals in e-commerce, mobile applications, cloud computing, digital transformation, data analytics, Testing, enterprise application integration and enterprise resource planning, with more than 307 active clients and 43 offices in over 18 countries, as of 31 March 2019.

History:

In August 1999, Mindtree Consulting Private Limited was founded by ten IT professionals, three of which invested through an entity incorporated in Mauritius. It was funded by the venture capital firms Walden International and Sivan Securities and received further funding in 2001 from the Capital Group and Franklin Templeton.

It became a public company on 12 December 2006 and was listed on the Bombay Stock Exchange and National Stock Exchange. Its IPO debuted on 9 February 2007 and closed on 14 February 2007. The IPO was oversubscribed by more than a hundred times. Mindtree announced a new brand identity and logo, with the slogan "Welcome to possible" on 28 September 2012.

In 2012, Mindtree setup its first U.S. delivery center (USDC) in Gainesville, Florida,

under the leadership of Scott Staples, co-founder, and Global Head of Sales. As of 2017, the company has 43 offices in over 17 countries.

Larsen & Toubro (L&T), an infrastructure major and one of the largest conglomerates in India, took over control of Mindtree in June 2019 and currently has a 61.08% stake in the company. L&T's takeover of Mindtree was described at the time as the first hostile takeover in the Indian IT industry.

In March 2020, Mindtree announced the appointment of Dayapatra Nevatia as COO with immediate effect. He joins the company from Accenture where he was the managing director as well as the director for delivery for advanced technology centers in India.

In April 2022, speculation arose that Mindtree and Larsen & Toubro Infotech will be merged.

Mindtree Merger:

In May 2022, it was announced that Mindtree, which was the other IT services subsidiary of Larsen & Toubro, will be merged into LTI, and the company will be renamed as LTIMindtree. Larsen & Toubro Group chairman A. M. Naik commented that there was not much overlap between the two companies in terms of industries and clientele; LTI was present in oil and gas sector, while Mindtree had a strong presence in sectors such as travel and hospitality.

The L&T Group has announced operationalization of the merger of Mindtree with LTI (Larsen & Toubro Infotech), effective November 14. The new entity, LTI Mindtree, will begin trading from November 24. Mindtree shareholders will get 73 LTI shares for every 100 shares they hold post-merger. LTI and Mindtree merger was announced in May 2022. The parent company, L&T, will hold 68.73% of the merged entity.

Services:

- Mindtree works in Application Development and Maintenance, Data Analytics,
- Digital Services, Enterprise Application Integration and Business Process Management, Engineering R&D, Enterprise Application Services, Testing, and Infrastructure Management Services.
- The company offers various research and development services including Bluetooth Solutions, Digital Video Surveillance, an integrated test methodology called MindTest, an IT infrastructure management and service platform called MWatch, the application management service, Atlas, SAP Insurance and Omnichannel.
- Mindtree's business is structured around clients in verticals such as Banking, Capital
- Markets, Consumer Devices & Electronics, Consumer Packed Goods, Independent Software Vendors, Manufacturing, Insurance, Media & Entertainment, Retail, Semiconductors and the Travel and Hospitality industry.
- Mindtree's workforce includes thousands of employees who serve as Microsoft Azure outsourced support agents.

Employees:

Mindtree has a total of 23,814 employees as of March 2021, of which 32% were women. Its workforce consists of employees of over 80 nationalities working from various offices around the globe. Out of its total workforce, 95% are software professionals and the remaining 5% work in support and sales.

LTI Mindtree will have 90,000 employees on its rolls, taking the overall headcount of the software businesses to 1.2 lakh, which includes that of LTTS, spread across in over 30 countries across five continents.

Subsidiaries:

Name	Location
Mindtree Software (Shanghai) Co. Ltd ^[44]	China
Bluefin Solutions Limited	United Kingdom
Bluefin Solutions Inc.	United States
Bluefin Solutions Sdn Bhd	Malaysia
Bluefin Solutions Pte Ltd	Singapore

Table 1.1: Company Location

Philanthropy:

Mindtree Foundation is a unit of Mindtree that works towards improving the lives of people with disabilities and the enhancement in the quality of primary education. Mindtree Foundation was incorporated on 20 November 2007, under section 25 of Companies Act. Mindtree's employees, assistive technologies and associations with NGOs led to the following:

- The launch of 'Udaan', a scholarship program to support the medical education of underprivileged students in association with Narayana Hrudayalaya Charitable Trust.
- The launch of 'I Got Garbage', a cloud-based platform aimed to simplify waste management and transform every waste picker in Bangalore, India, into an entrepreneur through a structured and governed waste management framework.

Chapter 2: Introduction of The Learning Program (IGNITE)

Offer Letter

Campus Onboarding <Campus.Onboarding@ltimindtree.com>

Tue, Feb 28,

2:32 PM

to me

Dear **Atul Kumar**,

We are pleased to offer you an internship opportunity (“**Internship**”) at LTIMindtree Limited (“**LTIMindtree**”). The terms and conditions of this offer are as follows and your Internship will be subject to and governed by these terms and conditions which shall be binding upon you receiving this email.

- | | | |
|-------------------------------|----------|------------------------------------|
| 1. Internship Duration | : | 9 to 12 weeks |
| 2. Commencement Date | : | 21 Feb 2023 |
| 3. Location | : | Remote |
| 4. Stipend | : | Milestone based - Refer 12. |

Additional Terms:

1. You acknowledge that the purpose of this Internship is to provide you with the opportunity to learn generally about information technology work and to gain practical experience and insights into the workplace and industry, and LTIMindtree does not derive an immediate advantage from the activities performed by you during the Internship Duration. It is essential for you to actively participate in this internship by completing the learnings successfully in the stipulated time.
2. The Stipend stipulated above is payable on completion of the milestones as set out in Clause 12 and on you joining LTIMindtree as a full-time employee. The amount of stipend that you will be entitled to will be determined as per the criteria set out

in Clause 12. Please note that the said amount will be a one-time payment and will not form part of your cost to the company as a full-time employee. The said amount will be payable to you subject to applicable statutory and other deductions, and applicable tax withholdings. Any costs and expenses borne by you in connection with the Internship shall be your sole responsibility.

3. You may be permitted to be absent during the Internship Duration after obtaining prior approval from your Mentor and your stipend may be reduced, at LTIMindtree's sole discretion and option, by the period of your absence. Prolonged, frequent, or unapproved absences may lead to immediate termination of your Internship upon notice from LTIMindtree.
4. While with us, you will be required to adhere to policies/practices of LTIMindtree as applicable to you in your capacity as an intern and as amended from time to time solely at the discretion of LTIMindtree ("**Policies**"). These policies will be shared with you before your internship commences and during the Internship Duration.
5. **Confidentiality:** As an intern, you will be privy to, have access to or receive Confidential Information (as defined below). You shall (i) use such Confidential Information solely in relation to and to fulfill your Internship; (ii) disclose Confidential Information only to such persons and as permitted in writing by LTIMindtree; (iii) treat the Confidential Information with all reasonable care; and (iv) return all Confidential Information (and all copies thereof) to LTIMindtree immediately upon termination or completion of your Internship. Your obligations to maintain secrecy and confidentiality of the Confidential Information shall continue after the termination of your Internship with LTIMindtree.

*"**Confidential Information**" which means any information, data or non-public business, commercial, personal, or technical information of LTIMindtree, its affiliates, parent company, their personnel, or that of their clients including but not limited to research and development projects, services, and business operations, which may be disclosed in writing, orally, electronically, by or on behalf of LTIMindtree. Any documents and information, which reflect, incorporate, and/or are generated using any such Confidential Information, will also be*

deemed as Confidential Information. All Confidential Information shall be deemed as LTIMindtree's trade secrets.

6. **Intellectual Property:** Title, interest, and ownership in all information, data, outputs, reports, codes, proprietary information or rights, materials, tools presentations, records, and intellectual property rights conceived, created, or developed by you in connection with or arising from your Internship, and/or making use of the Confidential Information shall vest solely and exclusively with LTIMindtree immediately upon creation without the need for any further act or payment of any remuneration. It is clarified that Section 19(4) of the Indian Copyright Act, 1957 shall not apply to any assignment of copyrights under this Letter and you hereby agree not to raise and waive all rights to raise, any objection or claim before the Indian Copyright Board with respect to the assignment pursuant to Section 19A of the Indian Copyright Act, 1957. Also, you may conceive newer and advanced methods to improve processes or systems during your internship; this will remain the sole property of LTIMindtree.
7. You agree to defend, indemnify, and hold harmless LTIMindtree for any loss, liability, claim, costs, fines, and or damage suffered by LTIMindtree and its personnel as a consequence of any breach by you of this Letter, LTIMindtree's instructions or any Policies.
8. LTIMindtree, at its sole discretion and option, reserves the right to withdraw, suspend and/or amend the offer of Internship and the terms of this Letter at any time prior to the Commencement Date specified above, and you acknowledge and agree that any such action and/or amendment by LTIMindtree shall be binding upon you immediately without any consequence on LTIMindtree.
9. You will be an Intern for the Internship Duration. This Letter and the Internship Duration may be suspended, terminated, or reduced (as appropriate) immediately with notice from LTIMindtree to you.
10. Issuance of Internship Certificate is always subject to the successful completion of the entire Internship Duration and at the sole discretion of LTIMindtree.

11. LTIMindtree may receive and collect personal data relating to you, including sensitive personal data or information (as defined in the Information Technology Act 2000 and rules made thereunder) (collectively “**Personal Information**”). LTIMindtree may process such Personal Information for relevant and limited purposes in connection with managing your internship and/or the business of LTIMindtree. You consent to (i) the collection, use, processing, storage, export, and transfer of your Personal Information by LTIMindtree and third parties; and (ii) the transfer and disclosure of your Personal Information held by LTIMindtree to any third parties within India or outside of India, in accordance with the LTIMindtree’s privacy policy and subject to applicable law.

12.

Internship Milestone	8 th Sem Stipend Reward (INR)*
Learning Completed with score 80% and above in first attempt in all 3 milestones	20,000
Learning Completed with score 60% - 79%	10,000
Learning Completed with score <60%	Not Entitled

Table 2.1: Stipend Table

*** This will be a one-time payment credited upon successfully completing the internship and post-onboarding as a full-time employee, followed by completion of L&D’s Initial Learning Program.**

This Letter contains the entire understanding between you and LTIMindtree for your Internship and supersedes all previous discussions and agreements, whether oral or otherwise.

You agree and acknowledge that the Internship is being granted solely for training purposes and that you are not an 'employee' or a 'workman' of LTIMindtree for the purposes of any employment statute or under any law, and you are not entitled to any wages or any employment benefits

(Including but not limited to leave and statutory benefits) that are provided solely to employees of LTIMindtree.

This Letter and the relationship between us shall be governed by the laws of India and the courts at Bangalore, Karnataka, shall have exclusive jurisdiction over any disputes that may arise therefrom. LTIMindtree may apply for injunctive or other appropriate relief from any court of competent jurisdiction.

You agree that your electronic signature below will have the same force and validity as a handwritten signature, and that your signature represents your acceptance of this Letter and your agreement to abide by the terms herein.

We are confident that you will enjoy your experience with us and that the learning you derive will be mutually beneficial.

Looking forward to seeing your expertise in action soon!

Thanking You,

University Liaison & Early Career Engagement
LTIMindtree

Learning Platform

We were given a platform named **EDUTECH UPSKILLS** which consists of different weekly courses related to DOTNET. Internship duration was 9 weeks which included 6 weekly Assessments and 3 **MILESTONES ASSESSMENTS**. Basically, my role in the company is to learn dotnet with the use of courses and complete the activities which are available in the same platform.

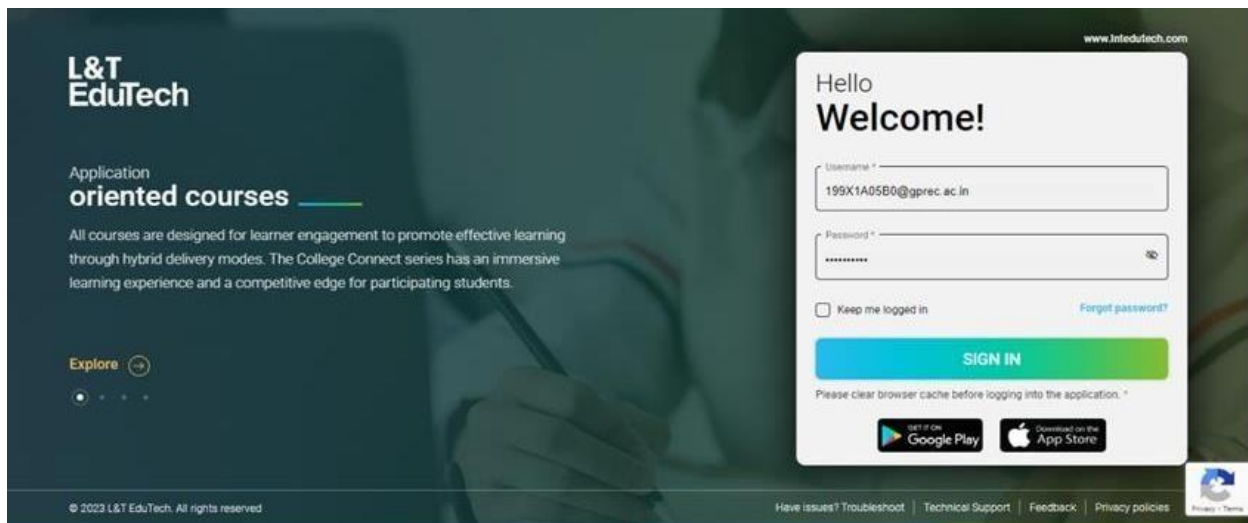


Fig 2.1: Upskill Login Page

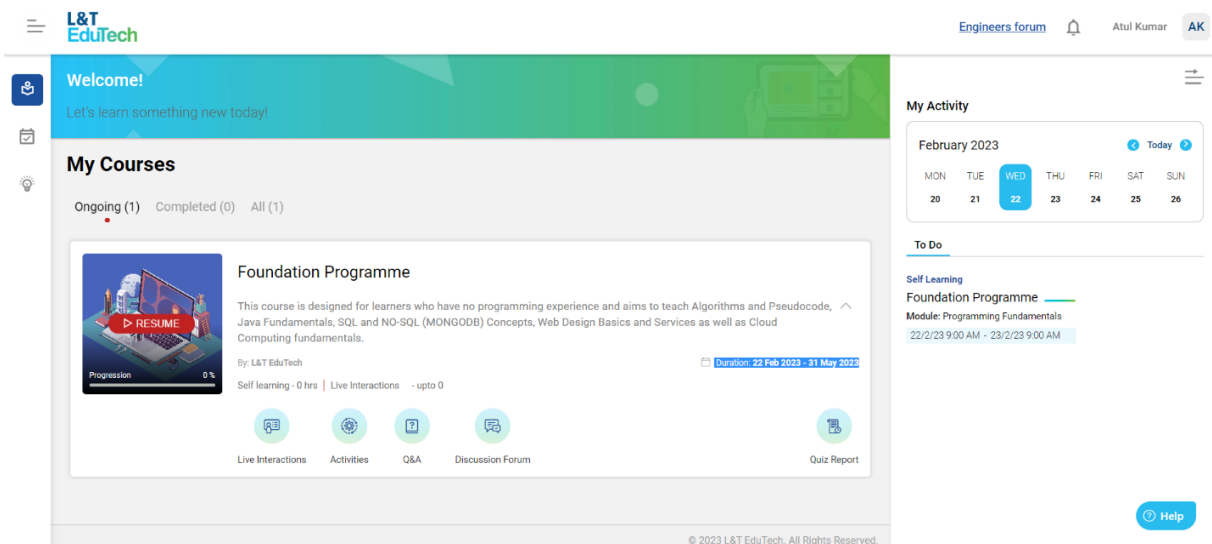


Fig 2.2: Upskill Portal

Domain: DotNet

.NET (pronounced "dot net") is a software framework developed by Microsoft that primarily runs on the Microsoft Windows operating system. It provides a programming model and a comprehensive set of libraries and tools for building various types of applications, including web applications, desktop applications, mobile applications, gaming applications, and more.

The .NET framework supports multiple programming languages, such as C#, Visual Basic .NET (VB.NET), F#, and C++/CLI (Common Language Infrastructure). It allows developers to write code using these languages and provides a runtime environment, known as the Common Language Runtime (CLR), which manages the execution of the code and provides features like memory management, exception handling, and security.

.NET also includes a large class library called the .NET Framework Class Library (FCL), which provides a wide range of pre-built functionality for common tasks such as file I/O, networking, database access, cryptography, and user interface development.

This library simplifies application development by offering reusable components and abstractions.

In addition to the traditional .NET framework, Microsoft has also introduced .NET Core, which is a cross-platform, open-source implementation of .NET. .NET Core allows developers to build applications that can run on Windows, macOS, and Linux operating systems. It provides a more modular and lightweight version of .NET, making it suitable for cloud-based and containerized environments.

Overall, .NET provides a powerful and versatile platform for developing a wide variety of applications, enabling developers to write code in different languages, access a rich set of libraries and tools, and target multiple operating systems.

Key Features of .NET

- **Language Independence:** .NET supports multiple programming languages, including C#, VB.NET, F#, and C++/CLI. Developers can choose the language they are most comfortable with or that best suits their project requirements.
- **Common Language Runtime (CLR):** The CLR is the execution engine of .NET. It provides services like memory management, garbage collection, code execution, and exception handling. The CLR ensures that code written in different .NET languages can be seamlessly executed together.
- **Framework Class Library (FCL):** The FCL is a comprehensive collection of reusable classes, types, and APIs that developers can leverage to build applications. It provides a wide range of functionality, including file I/O, networking, database access, XML processing, cryptography, and more. The FCL simplifies application development by offering pre-built components.
- **Object-Oriented Programming:** .NET follows the object-oriented programming (OOP) paradigm, allowing developers to create modular, maintainable, and extensible code. It supports features like classes, inheritance, polymorphism, and encapsulation, enabling developers to write efficient and scalable applications.
- **Interoperability:** .NET supports interoperability, allowing applications to communicate and interact with components and services built on different technologies. It integrates with existing systems and libraries through standards like COM (Component Object Model) interop, Platform Invoke (PInvoke), and Web Services.
- **Security:** .NET includes various security features to protect applications and data. It provides a sandboxed execution environment that isolates applications from each other, ensuring that they run securely. It also offers built-in security mechanisms for authentication, authorization, cryptography, and secure communication.
- **Cross-Platform Development:** With the introduction of .NET Core, .NET became cross-platform. Developers can create applications that run on Windows, macOS, and Linux.

operating systems. This allows for greater flexibility and the ability to target multiple platforms with a single codebase.

- **Development Tools and IDEs:** .NET provides a range of development tools and integrated development environments (IDEs) like Visual Studio, Visual Studio Code, and JetBrains Rider. These tools offer features like code editors, debugging tools, project management, and more, enhancing productivity and streamlining the development process.
- **Rich Ecosystem:** .NET has a vibrant ecosystem with a large community of developers, libraries, frameworks, and third-party tools. This ecosystem provides additional resources, documentation, support, and open-source projects that accelerate development and foster collaboration.

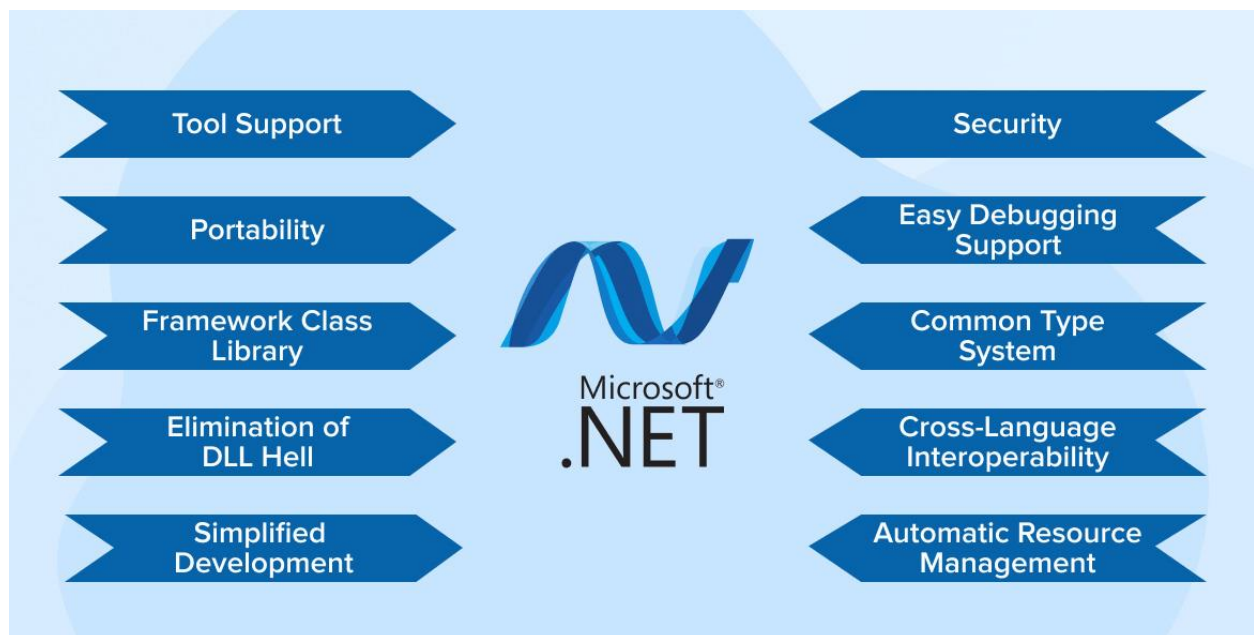


Fig 2.3: Features of .NET

Chapter 3: About Internship Work

Program

DOMAIN: DOTNET, DURATION: 9 WEEKS

WEEK 1:

- Core Programming Fundamentals
- Java Programming Fundamentals
- Looping Statement
- Arrays
- String, String builder, String buffer

WEEK 2:

- OOPS Basics
- Arrays as Object, Args and Return value
- Database Design
- SQL Querying
- Your First Queries
- Filtering Your Results
- Subqueries
- Grouping Your Data
- Joining Tables

WEEK 3:

- MongoDB Overview
- HTML, CSS, and JavaScript Basics
- Client-Server and Web Services Technology

- Cloud Computing

WEEK 4:

- .NET Framework
- Getting Started with Visual Studio Code
- Create Your First Program
- Debugging in Visual Studio Code
- Understanding C# Variables
- Understanding C# Operators
- Understanding C# Selection Statements
- Understanding C# Iterative Statements
- Understanding Debugging in C# .NET
- Understanding C# Class
- Understanding C# Class Fields
- Understanding C# Inheritance

WEEK 5:

- Understanding C# Polymorphism
- Understanding C# Interface
- C# Generics and Collections

WEEK 6:

- C# Delegates
- C# Events
- Features .NET
- Introduction to LINQ

- LINQ - Advance

WEEK 7:

- C# Features
- Exception Handling
- File Handling

WEEK 8:

- Coding Best Practices and Standards
- Multithreading
- JavaScript Introduction
- JavaScript Functions
- JavaScript Document Object model (DOM)

WEEK 9:

- TypeScript Introduction
- Variables and Data Types
- Functions
- Arrow Functions
- Classes
- Introduction to Angular
- Angular Component and Binding
- Angular Directive and Forms

Topic Covered Under DOTNET

- Foundation Programme
- Programming with C Sharp (C#)
- Typescript and Angular

Foundation Programme

In the foundation programme, we have learned Programming fundamentals using JAVA, OOP Basic concepts, SQL, JavaScript basics, and Cloud computing.

Programming fundamentals using Java:

Java is a popular programming language that was developed by Sun Microsystems (now owned by Oracle Corporation). It is known for its simplicity, platform independence, and extensive libraries and frameworks. Here are some fundamental concepts of Java programming:

1. **Syntax and Structure:** Java has a syntax similar to C and C++, which makes it easy for developers familiar with those languages to learn Java. Java programs are structured into classes, and each class consists of methods that contain the code to be executed.
2. **Variables and Data Types:** In Java, variables are used to store data. Each variable has a specific data type, such as int (integer), double (floating-point number), boolean (true or false), char (character), etc. Variables must be declared with a specific type before they can be used.
3. **Control Flow:** Java provides control flow statements to control the execution of code. These include conditional statements like if-else, switch-case, and looping statements like for, while, and do-while. These statements allow you to make decisions and repeat code based on certain conditions.

These are just a few fundamental concepts of Java programming. Java is a versatile language used in various domains, including web development, mobile app development (Android), enterprise software, and more. It continues to evolve with new features and enhancements while maintaining backward compatibility.

OOP Basic Concepts:

Object-Oriented Programming (OOP) is a programming paradigm that organizes code around the concept of objects, which are instances of classes. Here's an overview of The key principles and concepts of **OOP**:

1) Classes and Objects: In OOP, a class is a blueprint or template that defines the properties (attributes) and behaviors (methods) of objects. An object, on the other hand, is an instance of a class. For example, a class "Car" can define attributes like color, model, and methods like start(), stop().

2) Encapsulation: Encapsulation is the practice of bundling data and methods together within a class and providing access to the data through methods. It allows for data hiding, as the internal state of an object can only be accessed and modified through its defined methods, ensuring data integrity and security.

3) Inheritance: Inheritance is a mechanism that allows a class to inherit properties and behaviors from another class, known as the superclass or base class. The class that inherits from the superclass is called a subclass or derived class. Inheritance promotes code reuse and supports the "is-a" relationship. For example, a class "SUV" can inherit from the "Car" class, inheriting its attributes and methods.

4) Polymorphism: Polymorphism refers to the ability of objects of different classes to respond to the same message or method call in different ways. It allows for method overriding and method overloading. Method overriding occurs when a subclass provides its own implementation of a method defined in the superclass. Method overloading occurs when a class has multiple methods with the same name but different parameters.

5) Abstraction: Abstraction focuses on representing essential features without including the background details. It allows you to create abstract classes or interfaces that define a common set

of methods that concrete classes must implement. Abstract classes cannot be instantiated, but they can be used as base classes for other classes.

SQL(Structured Query Language):

SQL (Structured Query Language) is a programming language used for managing relational databases. It provides a standardized way to interact with databases, define and manipulate data, and perform various operations. Here are some key concepts and operations in SQL:

1. Database Creation:

- Use the ``CREATE DATABASE`` statement to create a new database.

2. Table Creation:

- Use the ``CREATE TABLE`` statement to define a new table within a database.
- Specify the table name and the names and data types of its columns.
- Optionally, define constraints such as primary keys, foreign keys, and uniqueness constraints.

3. Data Manipulation:

- Insert data into a table using the ``INSERT INTO`` statement.
- Update existing data using the ``UPDATE`` statement.
- Delete data using the ``DELETE FROM`` statement.

4. Querying Data:

- Retrieve data from one or more tables using the ``SELECT`` statement.
- Specify the columns to retrieve using the ``SELECT`` clause.
- Use the ``FROM`` clause to specify the table(s) from which to retrieve data.
- Apply conditions using the ``WHERE`` clause to filter data.
- Perform joins between tables using ``JOIN`` clauses based on common columns.

5. Filtering and Storing Data:

- Use the ``WHERE`` clause to apply conditions to filter data on specific criteria.
- Sort retrieved data in ascending or descending order using ``ORDER BY`` clause.

6. Aggregating Data:

- Perform aggregate functions like ``SUM``, ``AVG``, ``COUNT``, ``MIN``, and ``MAX`` to summarize data.

- Use the `GROUP BY` clause to group data based on one or more columns.
- Apply conditions on grouped data using the `HAVING` clause.

7. Joins:

- Combine data from multiple tables using different types of joins (`INNER JOIN`, `LEFT JOIN`, `RIGHT JOIN`, `FULL OUTER JOIN`).
- Specify join conditions using the `ON` clause.

These are some of the fundamental concepts and operations in SQL. SQL is widely used in database management systems like MySQL, Oracle, SQL Server, PostgreSQL, and others, and it plays a crucial role in data storage, retrieval, and manipulation within those systems.

Javascript Basic Concepts:

JavaScript is a programming language that adds interactivity to web pages. It is one of the three core technologies of the World Wide Web, along with HTML and CSS. JavaScript is used to create dynamic websites, interactive web applications, and games.

Here are some of the basic concepts of JavaScript:

- 1) Variables: Variables are used to store data in JavaScript. They can be declared and initialized using the var keyword.
- 2) Data types: JavaScript supports a variety of data types, including numbers, strings, Booleans, and objects. The data type of a variable can be determined using the typeof operator.
- 3) Operators: JavaScript operators are used to perform operations on data. Some of the most common operators include arithmetic operators, comparison operators, logical operators, and assignment operators.
- 4) Functions: Functions are blocks of code that can be reused. They are declared using the function keyword.
- 5) Events: Events are occurrences that happen on a web page, such as a user clicking a button or scrolling down the page. JavaScript can be used to respond to events using the addEventListener() method.

Cloud Computing:

Cloud computing is the delivery of computing services—including servers, storage, databases, networking, software, analytics, and intelligence—over the Internet (“the cloud”) to offer faster innovation, flexible resources, and economies of scale.

Here are some of the basic concepts of cloud computing:

- 1) Cloud computing services: Cloud computing services are offered in three main categories: infrastructure as a service (IaaS), platform as a service (PaaS), and software as a service (SaaS).
- 2) IaaS: IaaS provides the basic building blocks for cloud computing, such as virtual machines, storage, and networking.
- 3) PaaS: PaaS provides a platform for developing, deploying, and managing applications.
- 4) SaaS: SaaS provides software applications that are hosted and managed by a third-party provider.
- 5) Cloud deployment models: Cloud deployment models define how the cloud infrastructure is owned and managed. There are three main cloud deployment models: public clouds, private clouds, and hybrid clouds.
- 6) Public clouds: Public clouds are owned and operated by a third-party provider. They are the most common type of cloud deployment model.
- 7) Private clouds: Private clouds are owned and operated by an organization. They are typically used by organizations that have specific security or compliance requirements.
- 8) Hybrid clouds: Hybrid clouds combine public and private clouds. They are used by organizations that want to take advantage of the benefits of both public and private clouds.

Cloud computing offers a number of benefits, including:

- 1) Scalability: Cloud computing resources can be scaled up or down as needed. This makes it a good choice for businesses that experience fluctuating demand.
- 2) Cost savings: Cloud computing can help businesses save money on IT infrastructure costs.

3) Reliability: Cloud computing providers offer a high level of reliability. This is because their infrastructure is designed to be redundant and fault-tolerant.

4) Security: Cloud computing providers offer a variety of security features to protect data.

Programming with C#

C# (pronounced "C sharp") is a versatile programming language developed by Microsoft. It is part of the .NET framework and is widely used for developing a variety of applications, including web, desktop, mobile, and gaming applications. Here are some key features and concepts of C# which I have learned in this program:

1) Object-Oriented Programming (OOP):

- C# is an object-oriented language, which means it emphasizes the concept of objects, classes, and their interactions.
- It supports encapsulation, inheritance, and polymorphism, allowing for code reusability and modularity.

2) Syntax and Data Types:

- C# has a C-like syntax and is statically typed.
- It supports various built-in data types, including integers, floating-point numbers, characters, booleans, strings, and more.

3) Variables and Operators:

- Variables are declared using keywords like `var`, `int`, `string`, etc., and can be assigned values using the assignment operator (`=`).
- C# provides a wide range of operators for arithmetic, comparison, logical operations, and more.

4) Control Flow:

- Conditional statements: `if`, `else if`, `else` for executing different code blocks based on conditions.

- Loops: `for`, `while`, `do-while` for executing code repeatedly based on conditions.
- `switch` statement: for executing different code blocks based on different cases.
- C# provides exception handling mechanisms to handle runtime errors gracefully.
- The `try-catch` block is used to catch and handle exceptions, allowing for error recovery and program stability.

5) Generics:

- C# supports generics, allowing for type-safe code that can work with different data types.
- Generics provide reusable and flexible components by defining classes and methods with placeholder types.

C# is commonly used in conjunction with the .NET framework to develop a wide range of applications, including web applications using ASP.NET, desktop applications using Windows Forms or WPF, mobile applications using Xamarin, and game development using Unity. It offers a rich set of libraries and frameworks, making it a powerful language for building robust and scalable software solutions.

Typescript and Angular

In this module I have learned the basics of Typescript and Angular Framework.

TypeScript is a typed superset of JavaScript that adds static typing to the language. This means that the types of variables and expressions are known at compile time, which can help to prevent errors and improve code readability.

Here are some of the basic concepts of TypeScript:

- Variables: Variables in TypeScript can be declared with a type, such as number, string, or boolean. For example, the following code declares a variable named myname of type string.

let myName:string;

- Types: TypeScript supports a variety of types, including numbers, strings, Booleans, objects, arrays, and functions. The type of a variable can be determined using the `typeof` operator. For example, the following code determines that the type of the variable `myName` is a string.

typeof myName; //"string"

- Operators: TypeScript operators are used to perform operations on data. Some of the most common operators include arithmetic operators, comparison operators, logical operators, and assignment operators. For example, the following code uses the arithmetic operator `+` to add the numbers 1 and 2.

1+2; //3

- Functions: Functions in TypeScript can be declared with a type of signature, which specifies the types of the parameters and the return value. For example, the following code defines a function named `greet()` that takes a name as an argument and returns a string.

```
function greet(name:string):string{  
  return "Hello,"+name;  
}
```

- Classes: TypeScript supports classes, which are a way to create reusable code. Classes can be used to define objects, which are data structures that contain properties and methods. For example, the following code defines a class named `Person` that has a property named `name` and a method named `greet()`.

```
class Person{  
  name:string;  
  constructor(name:string){  
    this.name=name;  
  }  
  greet(){  
    return "Hello,"+this.name;  
  }  
}
```

Angular is a popular open-source framework for building web applications. Developed and maintained by Google, Angular combines declarative templates, dependency injection, end-to-end tooling, and other features to facilitate the development of robust and scalable applications. Here are some key aspects and features of Angular:

- **TypeScript:** Angular is built using TypeScript, a statically typed superset of JavaScript. TypeScript adds static typing, classes, modules, and other features to JavaScript, providing enhanced tooling and code quality.
- **Components and Templates:** Angular applications are based on components. A component consists of a TypeScript class that defines the component's behavior and a template that describes the component's UI. Templates use HTML combined with Angular-specific directives and bindings.
- **Directives:** Angular provides built-in directives, such as `ngIf`, `ngFor`, and `ngSwitch`, which enable dynamic rendering and manipulation of the DOM based on application state. Directives can be used to create custom behavior and reusable components.
- **Services and Dependency Injection:** Services in Angular are responsible for encapsulating logic that is independent of any specific component. Dependency Injection (DI) is a core feature of Angular that allows components and services to define their dependencies and have them automatically injected by the framework.
- **Routing:** Angular includes a powerful routing system that enables navigation between different views or components within an application. Routing allows for creating single-page applications (SPAs) and handling URL-based navigation.
- **Reactive Forms:** Angular provides robust support for building complex forms using reactive forms. Reactive forms offer a reactive and declarative approach to form handling, including validation, value changes tracking, and form submission.
- **Observables and RxJS:** Angular leverages the power of RxJS, a reactive programming library, to work with asynchronous data streams. Observables are used extensively in Angular to handle events, asynchronous operations, and data streams.
- **Testing:** Angular has built-in support for unit testing and end-to-end (E2E) testing. The framework provides tools and utilities for writing tests, including mocking dependencies and simulating user interactions.

- Angular CLI: The Angular Command Line Interface (CLI) is a powerful tool that simplifies the development process by generating boilerplate code, creating components, services, and modules, and providing build and deployment capabilities.
- Integration: Angular can be integrated with various third-party libraries and frameworks, such as Redux, NgRx, and Material UI, to enhance application functionality and UI design.

Program Completion

As the below picture shows I have completed all my learning with 100%

The criteria for writing the milestone test are to complete the learning at least 60% or more to participate in the test.

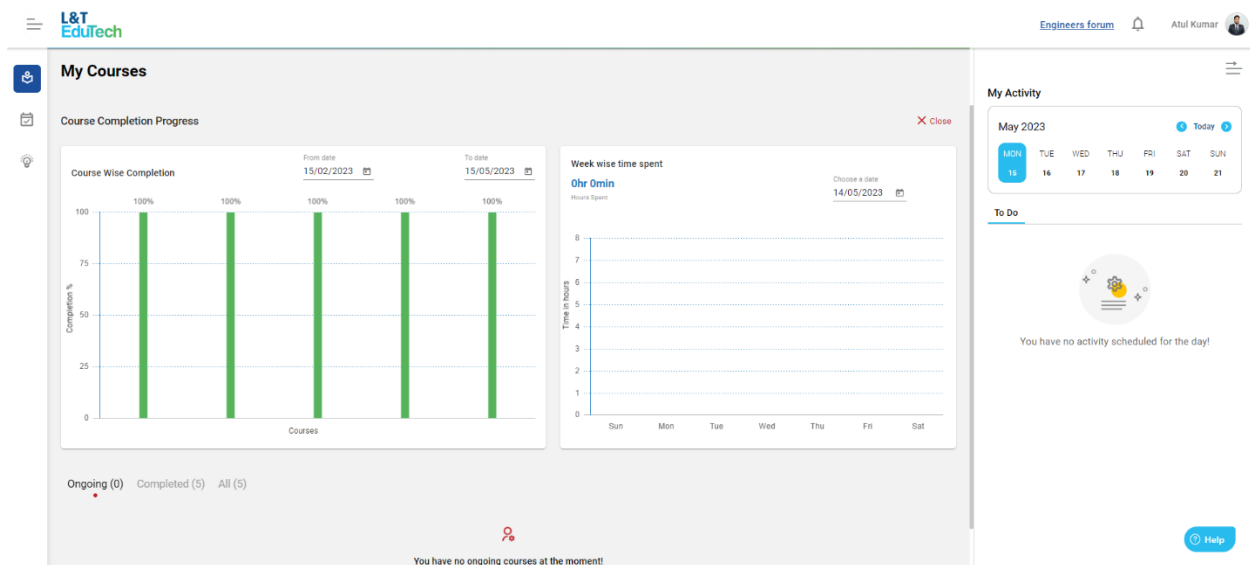


Fig 3.1: Course Completion Screenshot

Assessment Platform (NEOCODER)

NeoCoder provides training and upskilling for the technology teams with courses on the most in-demand technology topics, taught by the experts who know them best. Some key features include Hand-on training, Skill Assessment, Instructor-led training, Learning Paths, and analytics.

NeoCoder is the platform used to write tests in the IGNITE program. All tests like weekly and milestones were written on this platform.

The exam consists of 40% MCQ's and 60% coding questions.

Exam Duration is 2hr 45 minutes.

- 45 minutes for MCQ's
- 2 hours for Coding Questions
- Passing criteria is 60%

I have cleared all the 3 MILESTONES with more than 60%. I've also attached the screenshot for the same in the next chapter.

The screenshot displays the NeoCoder assessment platform interface. On the left is a dark sidebar with navigation icons for Courses, Contest, Open IDE, Marketplace, Leaderboard, and Assessments. The main content area has a top header with a search bar and user profile 'Atul'. Below this, a blue banner shows 'LTI_Ignite_Dotnet_Weekly Assessment_Week 04' with a 100% progress bar. A summary row lists: Start Date (25 Mar, 23), End Date (26 Mar, 23), Badges (0), Super Badges (0), and Tests (1). The main section is titled 'Weekly Assessment' with a 100% completion status. It lists the assessment 'LTI_Ignite_Dotnet_Weekly Assessment_Week 04' with start and end dates of 26 Mar 23. The 'Attempt' tab is active, showing 'Start Before: 26 Mar 23 | 11:45 AM (GMT +05:30)', 'Time Spent: 01:17:52', and 'Test Score: 72.00 / 75.00'. A table below shows scores for MCQ and Coding sections.

Sections	Score	Average Score	Top Score	Least Score
MCQ	27.00	21.24	29.00	0.00
Coding	45.00	26.54	45.00	0.00

Fig 3.2: Test Platform (NeoCoder)

SMEs/VILT Sessions

In the entire internship period, there is also a doubts clearing sessions called SME (Subject Matter Expert) or VILT (Virtual Interaction Lets Talk) which is held on twice a week on Wednesday and Friday. In these sessions they will solve our queries and show the things practically which we had learned through the self-learning program (videos).

Doubts clearing sessions have many benefits:

- **Enhanced Understanding:** Doubts clearing sessions provide an opportunity to seek clarification on topics or concepts that you find challenging or confusing. By having your doubts addressed by an expert or knowledgeable individual, you can gain a deeper understanding of the subject matter. This improved understanding can contribute to better academic or professional performance.
- **Personalized Learning:** Doubts clearing sessions often involve one-on-one interactions or small group discussions. This personalized approach allows you to receive individual attention and tailored explanations to address your specific doubts. It ensures that your unique learning needs are met and helps you progress at your own pace.
- **Increased Confidence:** When your doubts are resolved, it boosts your confidence in your abilities. You feel more assured in your knowledge and understanding of the topic. This increased confidence can positively impact your performance in exams, assignments, or work-related tasks, as you approach them with a stronger belief in your capabilities.
- **Improved Problem-Solving Skills:** Doubts clearing sessions often involve discussing specific problems or challenges related to the topic. By addressing these problems collaboratively, you can enhance your problem-solving skills. You learn different approaches, strategies, and techniques to tackle difficulties, which can be applied in various contexts.
- **Building Relationships:** Doubts clearing sessions provide an opportunity to interact with peers, teachers, mentors, or experts in the field. These sessions foster communication, collaboration, and the building of relationships within the learning or professional

community. Networking and establishing connections can be valuable for future learning, career opportunities, and personal growth.

- **Continuous Learning:** Doubts clearing sessions encourage a mindset of continuous learning. They emphasize the importance of seeking clarification, staying curious, and actively seeking solutions to problems. This mindset can extend beyond the session itself and inspire you to continue exploring and expanding your knowledge in other areas.

Overall, doubt clearing sessions offer a supportive environment to address questions, clarify doubts, and enhance learning. They contribute to improved understanding, increased confidence, and the development of critical thinking and problem-solving skills. Participating in such sessions promotes continuous learning and fosters connections within the learning or professional community.

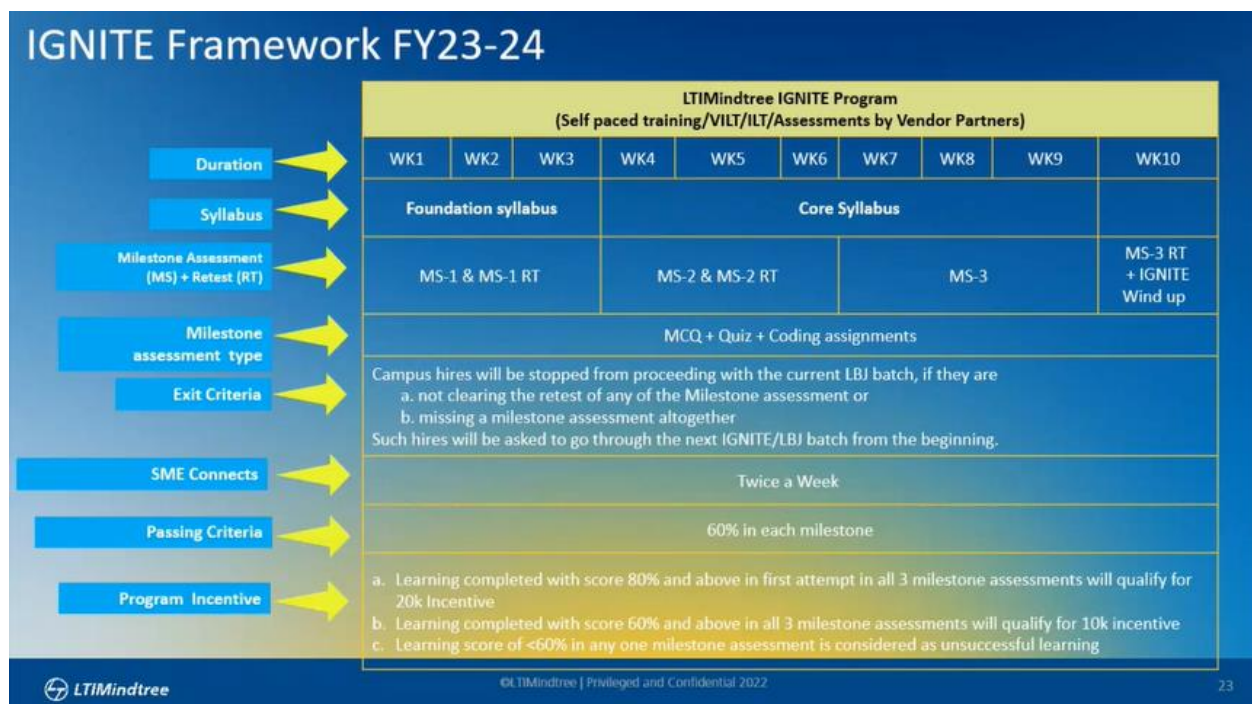


Fig 3.3: Internship Full Schedule

Chapter 4: Milestone Assessment Score (Result)

I have successfully completed my 9 weeks IGNITE program and cleared all three milestones which were conducted and mandatory to clear the Internship.

IGNITE Program Milestone Assessment Schedule

Milestone Assessment schedule for Java, .Net & Data track							
Date	Day	Assessment	Track 1 (Java)	Track 2 (.Net)	Track 3 (Data)	Duration	Timing
18-Mar-23	Saturday	Milestone 1	1-3 Week Syllabus	1-3 Week Syllabus	1-3 Week Syllabus	3 Hours	Virtual (3 slots)
08-Apr-23	Saturday	Milestone 2	4-6 Week Syllabus	4-6 Week Syllabus	4-6 Week Syllabus	3 Hours	Virtual (3 slots)
29-Apr-23	Saturday	Milestone 3	7-9 Week Syllabus	7-9 Week Syllabus	7-9 Week Syllabus	3 Hours	Virtual (3 slots)

Milestone Assessment schedule for Cloud track						
Date	Day	Assessment	Track 4 (Cloud)	Duration	Timing	
11-Mar-23	Saturday	Milestone 1	1-3 Week Syllabus	3 Hours	Virtual (3 slots)	
01-Apr-23	Saturday	Milestone 2	4-5 Week Syllabus	3 Hours	Virtual (3 slots)	
15-Apr-23	Saturday	Milestone 3	6-7 Week Syllabus	3 Hours	Virtual (3 slots)	

Note 1: If candidates are not able to clear the assessment in the first attempt, re-test will be scheduled after 2-3 days

Note 2: To enable the preparation for every milestone assessment, 2 weekly assessments are planned during the milestone duration on Saturdays with 1.5 Hours duration

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Fig 4.1: Milestone Test Schedule

Milestone 1

Test Score: 72.00/80.00

Percentage: 90%

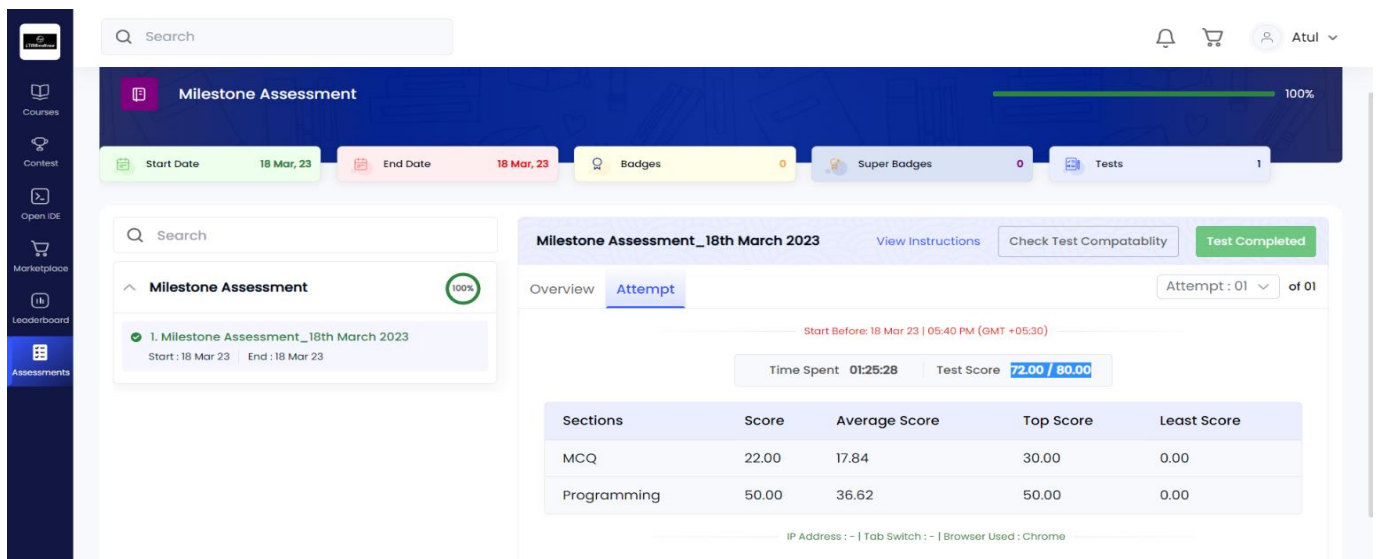


Fig 4.2: Milestone 1 Score Card

Milestone 2

Test Score: 88.00/100.00

Percentage: 88%

The screenshot displays the LTI Ignite Dotnet Milestone Assessment 02 Attempt 02 score card. The interface includes a sidebar with navigation links, a top search bar, and a main content area with a table of scores.

Search Bar: Search

Navigation Links: Courses, Content, Open ID, Marketplace, Leaderboard, Assessments

Assessment Details:

- Start Date: 12 Apr, 23
- End Date: 12 Apr, 23
- Badges: 0
- Super Badges: 0
- Tests: 1

Assessment Title: LTI_Ignite_Dotnet_Milestone Assessment 02_Attempt 02

Buttons: View Instructions, Check Test Compatibility, Test Completed

Overview / Attempt

Time Spent: 02:24:57

Test Score: 88.00 / 100.00

Sections	Score	Average Score	Top Score	Least Score
MCQ 01	16.00	14.68	20.00	0.00
MCQ 02	8.00	8.50	12.00	0.00
MCQ 03	4.00	5.60	8.00	0.00
COD 01	20.00	14.50	20.00	0.00
COD 02	40.00	23.76	40.00	0.00

Footer: IP Address: - | Tab Switch: - | Browser Used: Chrome

Fig 4.3: Milestone 2 Score Card

Milestone 3

Test Score: 87.00/100.00

Percentage: 87%

The screenshot displays a web application interface for a milestone score card. On the left is a dark blue sidebar with icons for Courses, Content, Open IDE, Marketplace, Leaderboard, and Assessments. The main content area has a dark green header for 'LTI_Mindtree_Milestone 03_Dotnet' with a 100% progress bar. Below the header, a summary bar shows 'Start Date: 27 Apr, 23', 'End Date: 1 May, 23', 'Badges: 0', 'Super Badges: 0', and 'Tests: 1'. A search bar is present above a list of milestones, with 'Milestone 03' selected and marked 100% complete. The main section, titled 'LTI_Mindtree_Ignite_DotNet_Milestone 03', includes buttons for 'View instructions', 'Check Test Compatibility', and 'Test Completed'. It shows 'Attempt: 01 of 01' and a 'Start Before' deadline of 01 May 23 at 11:15 AM (GMT +05:30). A summary bar indicates 'Time Spent: 02:31:32', 'Test Score: 87.00 / 100.00', 'Result: Pass', and 'Min. Pass Mark: 60.00'. A table provides a breakdown of scores by section.

Sections	Score	Average Score	Top Score	Least Score
MCQ 01	19.00	13.47	20.00	0.00
MCQ 02	12.00	10.08	12.00	0.00
MCQ 03	6.00	5.27	8.00	0.00
COD 01	10.00	14.72	20.00	0.00
COD 02	40.00	22.80	40.00	0.00

At the bottom, it shows 'IP Address: - | Tab Switch: - | Browser Used: Chrome'.

Fig 4.4: Milestone 3 Score Card

Conclusion and Future Perspective

The Internship period in LTIMindtree has been very productive and helpful. There is a big difference in the academic course and the tasks and activities during the actual work. In college we learn how quality is important and some basic quality software benefits and features, where during internship, it is learnt far more than the basics.

Alongside working on DOTNET I also learned how to use different tools like Git Hub , VS code, Oracle which makes the work very easy and effective. Lastly, the internship was a wonderful experience in the exposure to the professional field.

Conclusion:

Practical Experience: Through the internship, you gained hands-on experience working with Dotnet technology, allowing you to apply the theoretical knowledge you acquired during your studies. This practical exposure is crucial for understanding the real-world implementation of Dotnet and enhancing your technical skills.

Industry Relevance: Dotnet is a widely used technology in the software industry, and by interning in this field, you positioned yourself in a relevant and in-demand area. This experience will be highly valued by potential employers, as it demonstrates your ability to work with industry-standard tools and technologies.

Collaboration and Professional Development: During your internship, you likely had the opportunity to collaborate with other professionals, such as software engineers, project managers, and designers. This exposure helped you develop essential teamwork and communication skills, which are essential for a successful career in the tech industry.

Networking: Internships often provide valuable networking opportunities. You may have interacted with professionals, mentors, and colleagues who can serve as references or provide guidance and advice in the future. Building a strong professional network can be beneficial for your career advancement.

Future Perspective:

Job Opportunities: Completing an internship in Dotnet technology significantly increases your prospects of finding a job in the software development field. Dotnet is widely used across various industries, and there is a constant demand for skilled professionals in this domain. Your internship experience will give you a competitive edge when applying for job positions.

Skill Set Expansion: Dotnet is a vast ecosystem with numerous frameworks, libraries, and tools. Your internship might have focused on specific aspects of Dotnet, but it provides a solid foundation for further skill development. You can explore other Dotnet frameworks like ASP.NET, Xamarin, or Entity Framework, expanding your expertise and making you versatile in different areas.

Career Growth: Dotnet professionals have opportunities for growth and advancement in their careers. With experience gained during the internship, you can start as a junior developer and gradually progress to more senior roles such as software architect, technical lead, or project manager. Continuous learning and staying updated with the latest advancements in Dotnet will be key to your career growth.

Entrepreneurship and Freelancing: If you aspire to be an entrepreneur or a freelancer, your internship in Dotnet technology equips you with the skills to develop your own software solutions or offer your services to clients. You can leverage your internship experience to build a portfolio and showcase your expertise to potential clients or investors.

In conclusion, completing an internship in Dotnet technology provides you with practical experience, industry relevance, professional development, and networking opportunities. This internship serves as a steppingstone to future job prospects, skill set expansion, career growth, and entrepreneurial pursuits. By continuously learning and adapting to new trends, you can build a successful and rewarding career in the dynamic world of Dotnet technology.

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